

San Jacinto College

Snowflake Cortex Demo — Use Case Summaries

San Jacinto College serves ~31,800 credit students in the greater Houston area. This demo applies Snowflake Cortex AI to eight real institutional workflows — from advising and early alert to workforce outcomes and executive reporting.

1. Admissions & Enrollment Advising Chatbot

Cortex Search

Advisors spend significant time answering repetitive questions about admission requirements, deadlines, and procedures already documented in policy docs and handbooks.

Solution: Cortex Search over admissions handbooks, the program catalog, and financial aid policy surfaces instant, cited answers for advisors and students via a Streamlit chatbot — no RAG plumbing, no external vector DB.

Demo question: *"What are the TSI requirements for enrolling in College Algebra?"*

2. Student Success Early-Alert Analytics

Cortex Analyst

Early identification of at-risk students relies on manual grade and attendance review — slow, inconsistent, and dependent on SQL skills most advisors don't have.

Solution: Cortex Analyst over structured enrollment and course-attempt data lets advisors query student performance in plain English. SJC tracks A–C success (72% overall) and withdrawal rate (12%) as core strategic metrics.

Demo question: *"Which students in MATH 1314 have below a C midterm grade and attendance under 70%?"*

3. Stop-Out & Recruit-Back Identification

Cortex Analyst + Cortex Search

~6% of SJC's student population is in Stop-Out status (no enrollment in 3+ terms). Another ~2% are Recruit Backs — admitted but never registered. Manual re-engagement outreach is inconsistent.

Solution: Cortex Analyst over TargetX stage history surfaces prioritized re-engagement lists by program area and time-out-of-enrollment. Cortex Search over advising notes adds context on why a student stopped out.

Demo question: *"Show me Stop-Out students who were enrolled in Health Sciences and left within the last two years."*

4. Pathway Velocity & Completion Risk Scoring

Cortex Analyst + Snowpark ML

SJC tracks Pathway Velocity (SCH on-path ÷ SCH required) and Pathway Consistency (% of students staying in their declared program) as strategic metrics — but computing these at scale requires SQL expertise advisors don't have.

Solution: Cortex Analyst exposes pathway metrics in natural language. A Snowpark ML model scores completion risk and flags off-pace students for proactive intervention before they stop out.

Demo question: *"Which AAS students are completing less than 60% of their required pathway credits each semester?"*

5. CPD / Workforce Program Demand Analysis

Cortex Analyst + Cortex Search

San Jacinto's Continuing & Professional Development division serves thousands of non-credit learners across four centers (EDGE, BIT, Corporate, Community & Health). Program managers need enrollment trends and seat utilization without SQL.

Solution: Cortex Analyst over CPD enrollment data lets managers self-serve on program performance. Cortex Search over course descriptions helps match students to programs by industry or skill area.

Demo question: *"Which CPD programs have the highest growth rate over the last two years, and where are we hitting capacity?"*

6. Financial Aid Gap & Pell Eligibility Monitoring

Cortex Analyst + Cortex Search

56% of SJC students are Pell-eligible. Financial barriers are a leading cause of enrollment drop-off — students in Register Pending status with no aid package are at immediate risk of losing their seat.

Solution: Cortex Analyst over Banner aid and enrollment data identifies students with unmet need before payment deadlines. Cortex Search over financial aid policy docs answers student and advisor questions instantly.

Demo question: *"How many first-term students in Register Pending status have no financial aid package this semester?"*

7. Transfer Readiness & Workforce Outcomes Tracking

Cortex Analyst + Cortex Search

SJC reports Transfer Rate, Transferability, Employment Rate, and Earnings after Completion to THECB — but advisors can't easily access these metrics at the program or student level without running reports.

Solution: Cortex Analyst over graduate outcome data lets advisors and IR staff explore outcomes by program, campus, and cohort. Cortex Search over articulation agreements answers transfer credit questions in real time.

Demo question: *"What is the employment rate for Process Technology graduates compared to the state benchmark?"*

8. Institutional Research NL-to-SQL Dashboard

Cortex Analyst

SJC's IR team produces metrics for THECB, IPEDS, the Aspen Institute, and internal SLT reporting. Building and maintaining SQL reports for each audience is slow — and non-technical leadership can't self-serve.

Solution: Cortex Analyst connected to the full student data warehouse gives IR analysts and executives a conversational interface over enrollment, persistence, completion, and outcome data. Generated SQL is shown alongside results for full transparency and auditability.

Demo question: *"What was our Fall-to-Fall persistence rate for first-time-in-college students broken down by ethnicity and Pell status?"*

All Cortex Search, Cortex Analyst, and Streamlit assets are built in-platform via Snowflake Cortex Code. No external infrastructure required. All data remains inside the Snowflake perimeter.